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Microstructure and Mechanical Properties of New $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$ High-Entropy Alloys

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Effects of Co content on microstructures and hot hardness of a new high-entropy alloy system, $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$ ($x = 0.5$ to 2.0) were investigated. As cobalt content increases, the microstructure changes from dendrite to polygrain type and the constituent phases change from $\text{BCC} + \sigma$ at $x = 0.5$ to $\text{BCC} + \text{FCC} + \sigma$ at $x = 2.0$. The alloy hardness varies from Hv 788 at $x = 0.5$ to Hv 596 at $x = 2.0$. This can be explained with the relative amount of hard σ phase, medium hard BCC phase and soft FCC phase. All the $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$ alloys possess higher hardness level than that of Ni-based superalloys In 718/In 718 H from room temperature to 1273 K. They obey the Westbrook equation presenting the normal heating behavior. Both alloys of $x = 0.5$ and 1.0 exhibit a transition temperature higher than that of Co-based alloy T-800 by about 200 K. They also have a high hot hardness of Hv 347 at 1273 K, which is higher than those of In 718 and In718 H by Hv 220. The strengthening mechanism for their superiority is proposed. The $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$ alloy system has great potential in high-temperature applications.

Traditionally, alloy design for most alloys was based on one major element.^[1,2] Recently, a new alloy design concept utilizing high mixing entropy has been proposed by Yeh^[3] who defined a high-entropy alloy (hereafter called HEA) as having at least five major metal elements, each has a content higher than 5 at% but less than 35 at%. HEAs might exhibit attractive properties, such as high hardness, good thermal stability, high elevated-temperature strength, and good corrosion resistance explaining their potential applications in different aspects.^[4–9]

In this study, $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$ alloys (in molar ratio) are designed based on a previous alloy system of Al-Co-Cr-Cu-Fe-Ni, which has received significant

studies.^[6,10–12] Mo replaces Cu in these alloys in order to avoid the incipient melting point of Cu segregation along interdendritic regions in Al-Co-Cr-Cu-Fe-Ni system and to further increase the softening resistance at high temperatures. In addition, the Co content is varied to show a variety of microstructure and properties, which could enhance the understanding on the alloy system and broaden applications especially for high-temperature usage.

For assessing the strength, softening resistance, and wear resistance of the present alloys and commercial elevated-temperature alloys, in situ hot hardness measurement was performed in this study. Similar tests were also performed for tools, molds, and superalloys in literature.^[13,14] In analysis of hot hardness data, an equation form proposed by Westbrook is well known for identifying the softening resistance and revealing the softening mechanism.^[15]

$$H = Ae^{-BT} \quad (1)$$

where H = hardness in kg mm^{-2} ; T = test temperature in K; and A , B = constants. The constant A is called intrinsic hardness and B is softening coefficient or thermal coefficient of hardness. By this equation, two sets of A and B constants are commonly obtained at low- and high-temperature ranges corresponding to two different deformation mechanisms. At low-temperature region the major deformation mechanism of softening is slip, while at high-temperature region the major deformation mechanisms of softening are related with

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diffusional phenomenon, such as (a) slip along more than one slip system due to increased atomic mobility, (b) dislocation climb, and (c) grain boundary rotation.^[16,17]

Experimental

Table 1 shows the designated compositions of experimental $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$ alloys (Co- x alloy hereafter represent the alloy with cobalt content x). Raw elemental material, Al, Co, Cr, Fe, and Ni with purity higher than 99.5 wt% and master alloy, Fe-61.5 wt% Mo, all in granular shape were melted to prepare the alloys by using arc-melting method under a protective atmosphere of argon. Four times of re-melting and re-solidification were performed for each sample to assure the uniform composition. Microstructure and compositions were analyzed with scanning electron microscope (SEM, JEOL-5410) and attached X-ray energy dispersive spectrometer (SEM-EDS). Crystal structures were examined with X-Ray diffractometer (XRD, Rigaku ME510-FM2) at 30 kV and 20 mA with a scanning rate of 4° min^{-1} from 20 to 100° . Room-temperature Vickers hardness measurements were performed using a hardness tester (MATSUZAWA SEIKI MV-1) with a load of 5 kg and duration time of 15 s. Five points of indentation were taken to obtain an average hardness. Hot hardness measurement was conducted with a high-temperature hardness tester (Akashi AVK-CO) equipped with a heating and vacuum part (Akashi AVK-HF). A load of 5 kg and dwell time of 15 s were used. Before measuring, the chamber was purged with argon, and then evacuated to 1.9×10^{-4} torr for three times and finally filled with argon to a pressure close to 1 atm for the measurement. For each condition at least three indentations were taken to get an average value of hardness. Besides the measurement at room temperature, the sample was tested at temperatures from 573 to 1273 K at an interval of 100 K. Heating rate is 10 K min^{-1} for each temperature increase. The specimens for hot hardness measurements were 1.5 mm in thickness and 10 mm in diameter. They were ground and polished by standard metallographic procedure. Final polish was done with diamond paste of $1 \mu\text{m}$ grade.

Results and Discussion

Microstructure and Constituent Phases

Figure 1 shows the XRD patterns of $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$ alloys in the as-cast state. Only three crystal structures: FCC ($a = 3.60 \text{ \AA}$), BCC ($a = 2.87 \text{ \AA}$) with a small ordered peak at around 31° , and σ phase ($a = 8.94 \text{ \AA}$ and $c = 4.65 \text{ \AA}$) are

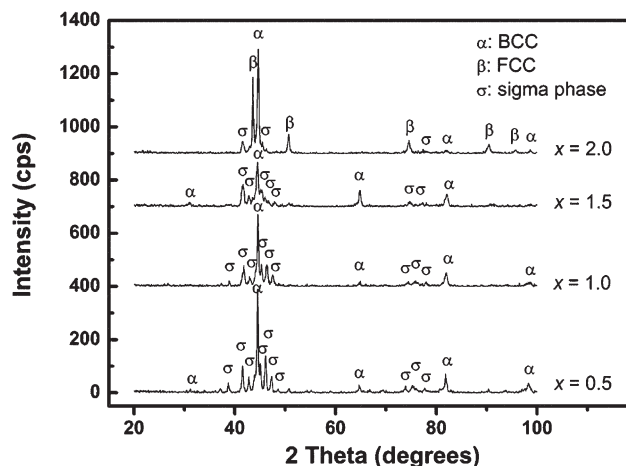


Fig. 1. XRD patterns of $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$ alloys with various Co contents.

identified. This reveals the interesting phenomena that although $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$ alloys have six principal elements, they exhibit a number of simple phases much less than seven phases allowed by Gibbs phase rule. This phenomenon has been generally found in published high-entropy alloys. It has been attributable to the chemical compatibility among constituent elements, and the high entropy effect interfering in the complexity of a multi-principal-element system to enhance the stability of solid solution phases.^[3–12]

Among the three phases of the present alloys, σ phase is in effect a solution-type intermediate phase. The structure of σ phase is similar to that of the binary intermediate phase CoCr , which is a body-centered tetragonal structure with lattice constants of $a = 8.81 \text{ \AA}$ and $c = 4.56 \text{ \AA}$ according to the JCPDS card. Related binary and ternary phase diagrams, i.e., Co-Cr, Fe-Cr, Ni-Mo, Fe-Mo, Fe-Cr-Mo, Cr-Mo-Ni, Co-Fe-Mo, Fe-Co-Cr, and Al-Mo-Ni^[18] show that the σ phase effectively dissolves Co, Cr, Fe, Ni, and Mo to a significant extent but does Al to a much lesser extent. For such a multi-element σ phase, Co, Fe, and Ni occupy the same site whereas Cr and Mo occupy the other same site in the crystal structure. Based on this, the present multi-element σ phase is in fact a solution phase of CoCr , FeCr , NiMo , and FeMo intermediate phases. From the practical viewpoint such a multi-element σ phase is very probable to have improved hardness and toughness since more kinds of solute atoms could increase the lattice distortion to enhance solution hardening, and also reduce the degree of long-range order to enhance toughness. In addition, multi-element σ phase could be fine in morphology due to the sluggish diffusion of atoms in the multi-element matrix.^[3,4] This could further improve hardness and toughness.

The Co-0.5, Co-1.0, and Co-1.5 alloys are characterized as having BCC and σ phases. From the relative peak intensities^[19] of BCC to σ phase, the volume fraction of BCC phase increases whereas that of σ phase decreases as the Co content increases from $x = 0.5$ to $x = 1.5$. The Co-2.0 alloy further contains a FCC phase indicating that the element Co can suppress σ phase and promote FCC phase.^[11,20] This is in consistence with the fact that Co is an FCC stabilizer.

Table 1. The compositions of the experimental alloys used in this study.

Composition	Designation	Crystal structure
$\text{AlCo}_{0.5}\text{CrFeMo}_{0.5}\text{Ni}$	Co-0.5	BCC + σ
$\text{AlCo}_{1.0}\text{CrFeMo}_{0.5}\text{Ni}$	Co-1.0	BCC + σ
$\text{AlCo}_{1.5}\text{CrFeMo}_{0.5}\text{Ni}$	Co-1.5	BCC + σ
$\text{AlCo}_{2.0}\text{CrFeMo}_{0.5}\text{Ni}$	Co-2.0	BCC + FCC + σ

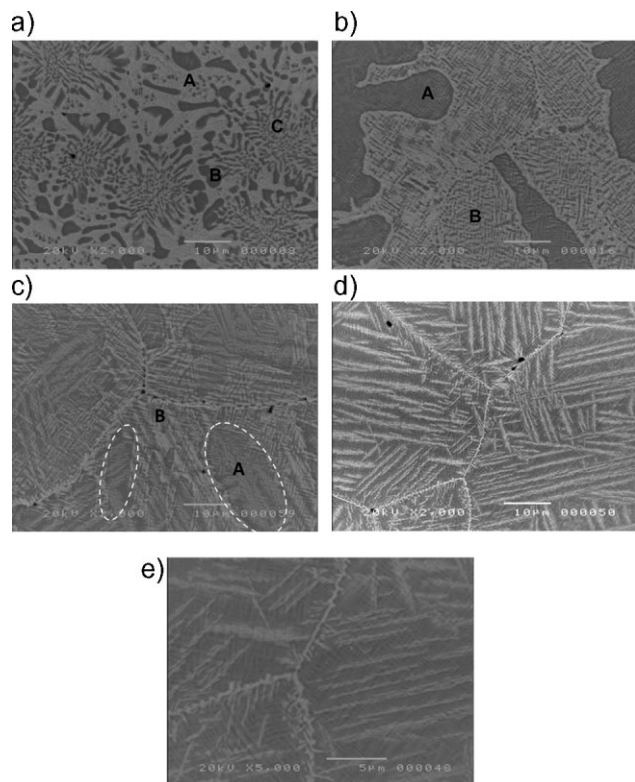


Fig. 2. SEM backscattered electron images of as-cast $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$ alloys: $x =$ (a) 0.5, (b) 1.0, (c) 1.5, (d) 2.0, and (e) 2.0 in higher magnification.

Figure 2 shows the SEM backscattered electron images of the as-cast $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$ alloys. The chemical compositions of different regions and phases were obtained by EDS and listed in Table 2. Figure 2(a) shows the microstructure of the Co-0.5 alloy, which has dendrite (labeled as A), interdendrite (labeled as B), and eutectic structure (labeled as

C). The white phases in the dendrite and eutectic structure are similar in composition which shows Cr and Mo contents higher than those in the overall composition. On the other hand, the fine gray phase inside the dendrite, the coarse gray interdendrite phase, and the gray one in the eutectic structure are all rich in Al and Ni, among which the gray interdendrite phase is even richer in Al and Ni than other gray phase.

From the XRD and composition analysis of constituent phases above, it leads to that the Al-rich dendrite of Co-0.5 alloy is a BCC phase in the beginning and then transformed into Cr- and Mo-rich white σ phase after precipitating out fine gray Al- and Ni-rich BCC phases (0.5 to 0.7 μm in size) during the cooling stage. The gray interdendrite with even higher Al content is the ordered BCC phase having small XRD ordered peak at around 31° since Al is the strong BCC former producing strong covalent $p-d$ hybridization with transition metals, e.g., $\text{Ni}_2\text{d-Alp}$ bonding. The eutectic structure is composed of gray BCC phase and σ phase in which gray BCC phase is a disordered one because it contains lower Al content.

The Co-1.0 and Co-1.5 alloys contain no eutectic regions but exhibit a simple dendritic structure composing of dendrite (labeled as A) and interdendrite (labeled as B). It can be seen that both dendrite and interdendrite display modulated structure like the spinodal decomposition structure as reported for $\text{Al}_x\text{CoCrCuFeNi}$ alloys.^[12] The characteristic width and spacing in the modulated structure are in the order of submicron meter. Table 2 shows that the overall composition of dendrites is rich in Al and Ni whereas that of the interdendrites is rich in Cr and Mo. This suggests that the Al- and Ni-rich dendrite with BCC structure first forms and then due to the partitioning of more Cr and Mo to liquid phase upon subsequent cooling, Cr- and Mo-rich interdendrite with BCC structure forms as the outer envelope of the cored

Table 2. Chemical compositions of different regions and phases in $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$ alloys.

x-Value	Region or phase	Al	Cr	Fe	Co	Ni	Mo
0.5	Nominal	20.0	20.0	20.0	10.0	20.0	10.0
	DR ^[a] (A region overall)	17.9	21.7	20.5	9.9	17.6	12.4
	A region (white phase)	11.9	26.8	23.6	9.7	12.3	15.7
	A region (gray phase) ^[c]	24.7	15.6	17.2	10.3	23.8	8.4
	ID ^[b] (B region)	34.2	8.8	13.2	9.3	31.9	2.6
	C region (gray phase) ^[c]	26.4	16.2	16.9	9.3	23.2	8.0
	C region (white phase) ^[c]	13.8	25.9	23.4	9.4	13.3	14.2
	Nominal	18.2	18.2	18.2	18.2	18.2	9.0
1.0	DR ^[a] (A region overall)	24.5	12.3	15.6	18.1	24.4	5.1
	ID ^[b] (B region overall)	11.3	21.0	19.3	17.5	14.1	16.8
	Nominal	16.7	16.7	16.7	25.0	16.7	8.2
1.5	DR ^[a] (A region overall)	18.3	15.3	15.8	24.2	18.0	8.4
	ID ^[b] (B region overall)	14.9	17.7	16.9	24.5	15.5	10.5
	Nominal	15.4	15.4	15.4	30.8	15.4	7.6
2.0	Grain (overall)	15.9	15.4	15.1	29.6	15.3	8.7
	Grain (white phase) ^[c]	16.3	15.0	14.3	28.1	13.7	12.6
	Grain (gray phase) ^[c]	14.8	15.7	16.7	31.8	15.1	5.9
	Grain boundary ^[c]	12.6	16.4	13.7	27.7	12.6	17.0
	Nominal	15.4	15.4	15.4	30.8	15.4	7.6

[a] DR: dendrite.

[b] ID: interdendrite.

[c] Values are approximate because the phase or region size detected is below the instrument resolution, i.e., $<1 \mu\text{m}$.

dendrite. Because of the difference in Cr and Mo contents, the BCC dendrite precipitates out a smaller volume fraction of fine σ phase (white phases) but the BCC interdendrite precipitates out larger volume fraction of σ phase in the cooling stage after solidification. The precipitation results in modulated structures (white phase is σ and gray phase is BCC) both in dendrite and interdendrite.

Figures 2(d) and (e) show the microstructure of Co-2.0 alloy with two magnifications. The alloy displays a polygrain structure without dendritic features. This is like pure metals suggesting its melting point range is very narrow. The grain size of Co-2.0 alloy is around 200 μm . Grain boundaries have thin layer of white σ phase and grain interiors contain white large stringers around 50 μm in length and a modulated structure between large stringers. The large stringers are in fact composed of short rod-like particles ranging from submicron to micron meter in length. From XRD analysis, the modulated structure is composed of FCC and BCC since their XRD peak heights are similar.

Mechanical Properties of $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$ Alloys

Room-temperature Hardness and Crack Length of Indentation

Figure 3 show the room-temperature Vickers hardness and overall crack length around indentation of as-cast $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$ alloys as functions of Co content. Those of the $\text{Al}_x\text{CoCrCuFeNi}$ alloys^[6] from which the present alloy were derived are compared to see the property improvement. For the variation of hardness shown in Figure 3(a), it relates with the relative volume fractions of hard σ phase, medium hard BCC phase, and soft FCC phase. Evidences have shown that σ phase is the hardest phase since microhardness measurements on the dendrite (with less σ phase) and the interdendrite (with more σ phase) in the Co-1.0 alloy give Hv 634 and Hv 824, respectively. Therefore, the volume fraction of σ phase is the main factor in determining the hardness of these alloys. The alloy hardness decreases with decreased volume fraction of σ phase as the Co content increases from $x = 0.5$ to $x = 1.5$. Since Co-2.0 alloy contain a certain portion of soft FCC phase, a great hardness decrease occurs from Hv 711 at $x = 1.5$ to Hv 596 at $x = 2.0$. For the variation of crack length which is an indicator of fracture toughness,^[21] it can be found from Figure 3(a) that the toughness increases as hardness decreases. This follows the general trend in materials. However, it is noted that Co-0.5 and Co-1.0 alloy possess similar hardness but the crack length of Co-0.5 alloy is much longer than that of Co-1.0. This is attributable to their microstructure difference since Co-1.0 alloy has fine modulated microstructure in the dendrite and interdendrite which could substantially retard the crack propagation whereas Co-0.5 alloy has a continuous hard but brittle white σ phase as its matrix among which coarse BCC gray phase disperses.

From the comparison of the hardness and toughness levels between the present alloys and the base alloys, it can be seen that the present alloys are superior over the base alloy in the aspects of hardness and fracture toughness. For example, the

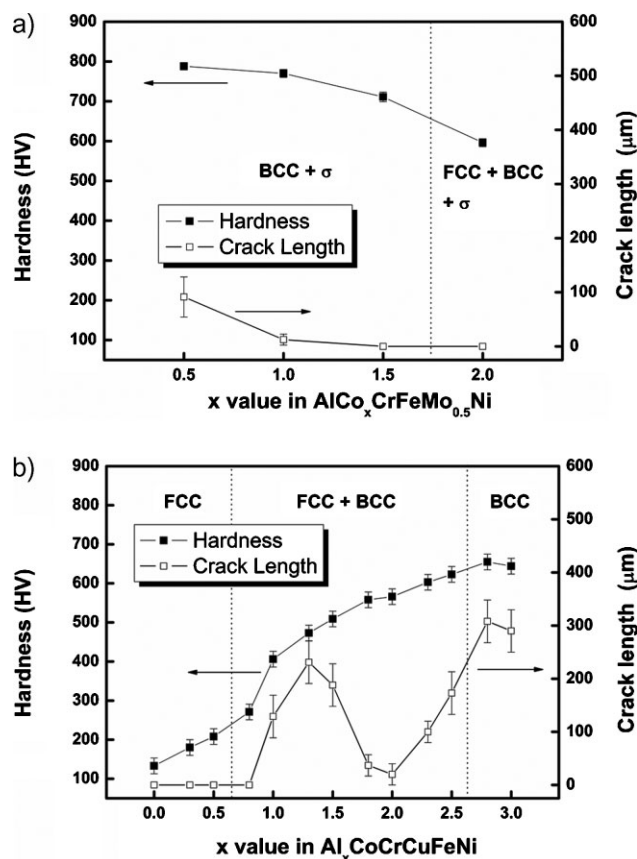


Fig. 3. (a) Vickers hardness and total crack length of $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$ alloys as functions of Co content; and (b) those of $\text{Al}_x\text{CoCrCuFeNi}$ alloys for comparison with (a)[6].

hardest Co-0.5 alloy has a crack length 91 μm similar to that of $\text{Al}_{1.0}\text{CoCrCuFeNi}$ and $\text{Al}_{2.3}\text{CoCrCuFeNi}$ alloys, but Co-0.5 alloy has a hardness of Hv 788 much higher than the latter two having Hv 406 and Hv 603, respectively. Furthermore, even the best alloy $\text{Al}_{2.0}\text{CoCrCuFeNi}$ having a crack length about 20 μm and a hardness of Hv 566 due to its modulated spinodal structure of B2 ordered BCC and BCC phases is still inferior to Co-1.0 alloy having a crack length about 13 μm and a hardness of Hv 770. In overall comparison, the hardness superiority is attributable to the high volume fraction of σ phase in the present alloys since σ phase is much harder than BCC phase which is the main strengthening phase in the base alloys. As for the toughness superiority, it is not clear about the exact mechanism, but it indeed suggests that a fine modulated structure of BCC and σ phases is better than the modulated spinodal structure of B2 and BCC phases.

Hot Hardness

Figure 4 shows the variations of hot hardness (H) with temperature (T) for the $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$ alloys, Ni-based superalloy of In 718 (as-received) and In 718 H (precipitation hardened), and Co-based superalloy, Tribaloy T-800 (data are obtained from Stellite Company website). The compositions of In 718 and T-800 are noted in Table 3. It can be seen that the curves of Co-0.5, Co-1.0 and T-800 alloys are almost parallel

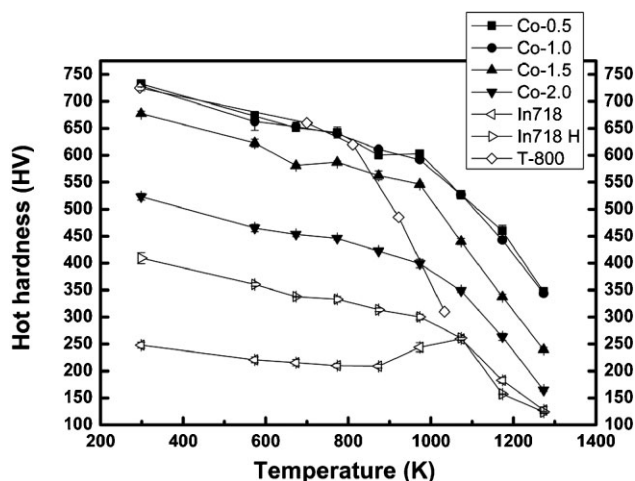


Fig. 4. Hot hardness vs. temperature plots for $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$ alloys ($x = 0.5, 1.0, 1.5$, and 2.0), In 718, In 718 H and T-800 alloys from room temperature to 1273 K.

from room temperature to 810 K. But at above 810 K the hardness value of T-800 decreases faster and becomes Hv 310 at 1033 K, which is much lower than that, around Hv 550, of Co-0.5 and Co-1.0 alloys. In addition, Co-0.5 and Co-1.0 alloys exhibit the highest hot hardness of about Hv 340 at 1273 K, which is much higher than that, around Hv 127, of In 718 and In 718 H. It is noted that In 718 is the only one to show the abnormal heating behavior (i.e., hardness increase with temperature increase) in the temperature range from 873 to 1073 K, which is mainly attributable to the precipitation hardening effect at this temperature range as evidenced by the precipitation hardening used to strengthen In 718 H. However, In 718 becomes soften at a faster rate at above 1073 K.

The above data of Figure 4 are re-plotted as Figure 5 by using $\ln(H)$ vs. T . The $\ln(H)$ versus T plots for $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$ alloys can be represented by two straight lines which intersect in the temperature range 950–1150 K, suggesting a change of deformation mechanism^[15] in this transition as mentioned in the Introduction section. The values of transition temperature (T_T), intrinsic hardness (A), and softening coefficient (B) calculated according to Westbrook^[15] are

summarized in Table 3. It can be seen that Co-0.5, Co-1.0, and Co-1.5 alloys have almost the same softening coefficient (B_I) up to the transition temperature (T_T). However, the slope (B_{II}) above transition temperature of Co-1.5 alloy becomes more negative than those of Co-0.5 and Co-1.0 alloys, indicating that the hardness of Co-1.5 alloy declines faster than Co-0.5 and Co-1.0. In contrast, Co-0.5 and Co-1.0 alloys have the least negative value of softening coefficient (B_{II}), and its value is about half that of In 718, suggesting that Co-0.5 and Co-1.0 alloys could sustain its strength up to 1273 K very effectively. Although Co-2.0 alloy possesses the most negative value of softening coefficient (B_{II}) among all alloys compared, indicating that Co-2.0 alloy has the weakest resistance against softening, it still has a higher hardness level than In 718, In 718 H over all temperature range and than that of T-800 at temperatures above 1000 K. In summary, the $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$ alloy system has great potential in high-temperature applications.

It is interesting to see that the softening coefficients of pure metals such as Mo, Ni, Cr, Si, and Fe^[22] are two order magnitudes larger than those of $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$ alloys, In 718, and T-800, indicating that alloying indeed has tremendous effect for lowering the softening coefficient. As $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$ contains a substantial volume fraction of σ phase, which decreases with increasing the Co content, it suggests that the σ phase is the main strengthener not only at room temperature but also at high temperatures. This is an important finding.

Besides the strengthening from σ phase mentioned above, multi-element solution hardening also has contributions to the strength and the resistance to softening. For the region below transition temperature where the deformation mechanism is slip, multi-element solid-solution phases such as BCC, FCC, and σ phase would have a larger resistance to slip deformation because of solution hardening mechanism. This is reasonable since, even at high temperatures, the matrix is still effectively strengthened by highly concentrated solute atoms according to the mechanism of solution hardening. For the region above transition temperature where the major deformation mechanism is diffusional phenomenon, multi-element solid-solution

Table 3. Transition temperature (T_T), intrinsic hardness (A) and softening coefficient (B) of $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$ alloys and compared superalloys^[15].

Composition	Transition temperature (T_T) K	Below T_T		Above T_T	
		A_I	B_I	A_{II}	B_{II}
Co-0.5	1003	801	−3.04E-4	3551	−1.79E-3
Co-1.0	995	796	−2.99E-4	3528	−1.82E-3
Co-1.5	988	741	−3.16E-4	8059	−2.74E-3
Co-2.0	1050	586	−3.79E-4	20343	−3.76E-3
T-800 ^[a]	818	791	−2.84E-4	8041	−3.12E-3
In 718 ^[b]	1069	268	−3.10E-4	11766	−3.55E-3
In 718 H ^[c]	1037	487	−5.29E-4	13778	−3.74E-3

[a] Composition of T-800: Co: 47.6%; Mo: 28%; Cr: 18%; Si: 3.4%; Fe: 1.5% and Ni: 1.5%.

[b] Composition of In 718: Ni: 53%; Cr: 19%; Fe: 18.5%; Nb: 5.1%; Mo: 3%; Ti: 0.9% and Al: 0.5%.

[c] Annealed at 1253 K for 30 min, air cooled, and aged at 1033 K for 8 h, furnace cooled at a rate of 55 K, heated to 893 K for 8 h, air cooled.

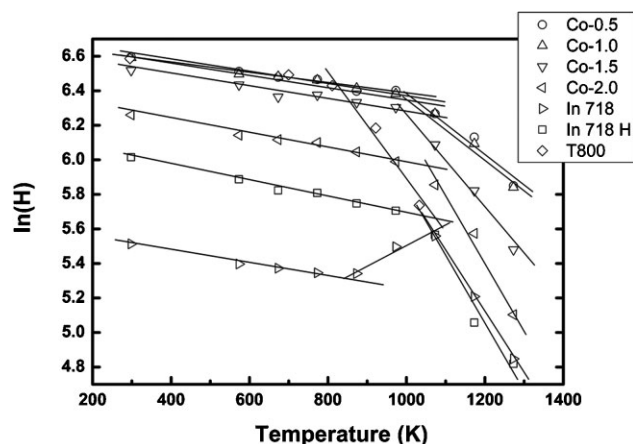


Fig. 5. $\ln(\text{hot hardness})$ plotted against temperature (from room temperature to 1273 K) for $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$, In718, In 718 H, and T-800 alloys (data are obtained from the Stellite Company website).

phases with highly concentrated solute atoms also have the benefit to resist creep deformation as the vacancy concentration in this high mixing entropy matrix is intrinsically lower than that of a pure metal matrix. This is because vacancy justifies its existence with increasing the mixing entropy but the justification would become less strong in the high mixing entropy matrix.

Conclusions

- i. With the increased content of Co, an FCC stabilizer, the crystal structure of $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$ HEAs gradually changes from BCC + σ phase at $x = 0.5$ to BCC + FCC + σ phase at $x = 2.0$.
- ii. The microstructure changes from dendrite + interdendrite + eutectic structure at $x = 0.5$ to dendritic structure at $x = 1.0$ and 1.5, and finally to a polygrain structure at $x = 2.0$. For the alloys of $x = 1.0$, 1.5, and, 2.0, modulated structure prevails in the dendrite, interdendrite, and polygrain.
- iii. Alloy of $x = 0.5$ is the hardest one due to its highest volume fraction of hard σ phase. As Co content increases, the alloy hardness decreases due to the decreased amount of σ phase.
- iv. $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$ ($x = 0.5$ to 2.0) alloys exhibit a superior combination of hardness and toughness over the base alloys they derived from, $\text{Al}_x\text{CoCrCuFeNi}$.
- v. All $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$ alloys possess higher values of hardness than that of Ni-based superalloys In 718/In 718 H at temperatures from room temperature to 1273 K.
- vi. Co-0.5 and Co-1.0 alloys have a transition temperature higher than that of T-800 by around 200 K. They also exhibit a hot hardness of around Hv 343 at 1273 K, which is much higher than Hv 127 of In 718. Co-0.5 alloy displays the lowest softening coefficient that is only half that of In 718.
- vii. The present $\text{AlCo}_x\text{CrFeMo}_{0.5}\text{Ni}$ alloys show good slip deformation resistance in the lower temperature regime

and creep deformation resistance in the higher temperature regime. Besides the main strengthening of hard σ phase, multi-element solution hardening and a lower vacancy concentration also have contributions.

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